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A Template-Based Website Builder

A project Submitted To Council of the College of Computer Science &
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in Information Systems

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(يَرْفَعُ اللَّهُ الَّذِينَ آمَنُوا مِنْكُمْ وَالَّذِينَ أُوتُوا الْعِلْمَ دَرَجَاتٍ)

[المجادلة: 11]



Dedication

To my beloved family. . .

My Mother

My Father

My Sisters

To my lovely friends

And everyone who cares . . .

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Librarians at university of Al Qadisiyah. I am also indebted to Information Systems Department faculty; my university also deserve special thanks for their assistance in supplying the relevant literatures.

Supervisor Certification

I certify that the project entitled “A Template-Based Website Builder” was prepared under my supervision at the Information Systems Department / College of Computer Science & Information Technology /University of Al Qadisiyah, as partial fulfillment of the requirements needed for award of the degree of Bachelor in Information Systems.

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Examination Committee Certification

I hereby certify that this project entitled "A Template-Based Website Builder" Was prepared under my supervision at the Information Systems Department / College of Computer Science & Information Technology /University of Al Qadisiyah, as a partial fulfillment of the requirements for the Bachelor in Information Systems.

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LIST OF ABBREVIATIONS

Abbreviation	Full Term
AI	Artificial Intelligenc
API	Application Programming Interface
BSON	Binary JavaScript Object Notation
CSS	Cascading Style Sheets
CORS	Cross-Origin Resource Sharing
DB	Database
DOM	Document Object Model
FPS	Frames Per Second
HTML	HyperText Markup Language
IDE	Integrated Development Environment
I/O	Input/Output
JSON	JavaScript Object Notation
JS	JavaScript
JWT	JSON Web Token
MERN	MongoDB, Express.js, React, Node.js
NoSQL	Not Only SQL (Non-relational database)
REST	Representational State Transfer
SEO	Search Engine Optimization
SPA	Single Page Application
SQL	Structured Query Language
TWB	Template Website Builder
UI	User Interface
UX	User Experience
VCS	Version Control System
VS Code	Visual Studio Code
WYSIWYG	What You See Is What You Get

Abstract

Establishing a robust and visually compelling web presence has become a fundamental necessity for businesses, creators, and individuals in the contemporary digital era. Despite this, web development remains a polarized field, often divided between professional coding environments that require deep technical expertise and traditional drag-and-drop builders that sacrifice creative freedom and performance for ease of use. This research project introduces the TWB — Template Website Builder, a pioneering platform designed to bridge this gap by offering a high-performance, WYSIWYG (What You See Is What You Get) workspace that prioritizes both user autonomy and technical excellence.

The centerpiece of the TWB project is its innovative Hybrid Positioning Engine. Unlike standard builders that confine users to rigid grid systems, TWB enables a seamless transition between freeform absolute positioning allowing for precise, pixel-perfect design and structured responsive flows that ensure cross-device compatibility. This approach provides the intuitive feel of professional graphic design tools while maintaining the structural integrity and performance of a modern website. Furthermore, the platform addresses the pervasive issue of "code bloat" by generating clean, optimized, and production-ready code in real-time.

Leveraging a robust technical stack comprising React.js, Node.js, and MongoDB, TWB implements advanced features such as an infinite canvas workspace and real-time interactive state previewing. The research concludes that by integrating these professional-grade design capabilities into a no-code environment, TWB successfully democratizes high-end web development. It provides an efficient solution for users seeking to create sophisticated, responsive, and performance-driven websites without the steep learning curve traditionally associated with professional web design tools.

Keywords: *Web Development, No-Code, Drag-and-Drop, WYSIWYG, Hybrid Positioning Engine, React.js, Responsive Design, Website Builder.*